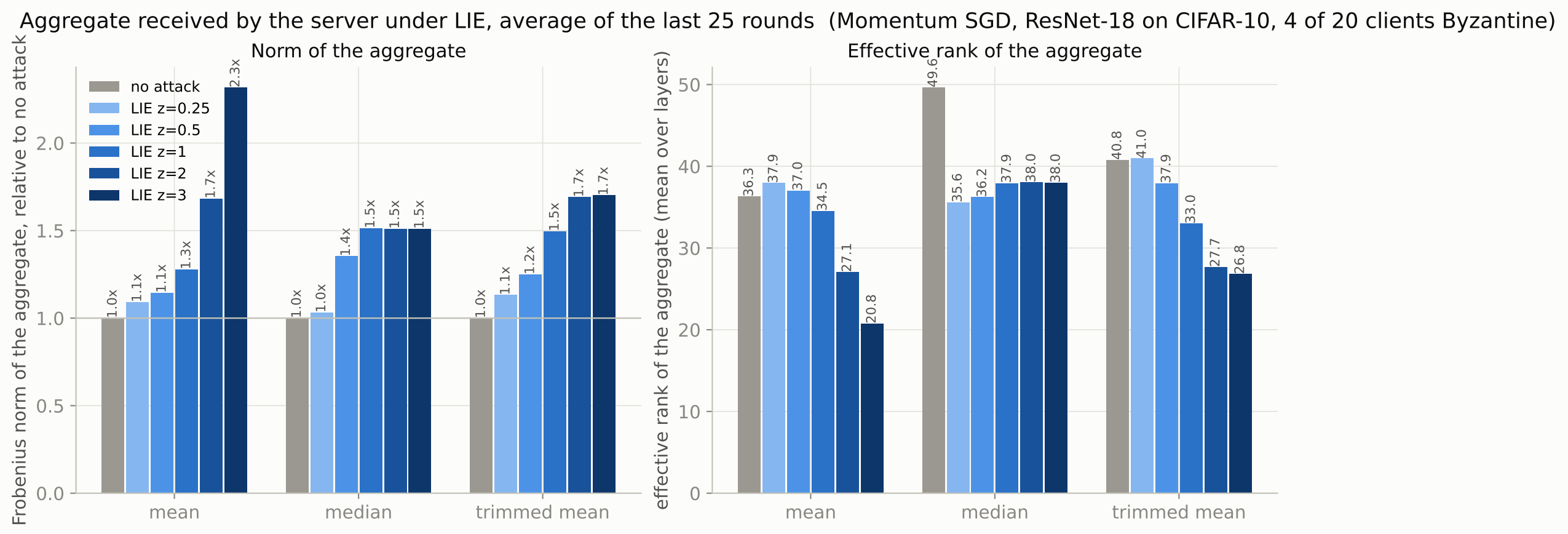
Aggregate received by the server under LIE, average of the last 25 rounds (Momentum SGD, ResNet-18 on CIFAR-10, 4 of 20 clients Byzantine)
Norm of the aggregate
Frobenius norm of the aggregate, relative to no attack
no attack
LIE z=0.25
LIE z=0.5
LIE z=1
LIE z=2
LIE z=3
1.0x
1.1x
1.1x
1.3x
1.7x
2.3x
1.0x
1.0x
1.4x
1.5x
1.5x
1.5x
1.0x
1.1x
1.2x
1.5x
1.7x
1.7x
0.0
0.5
1.0
1.5
2.0
mean
median
trimmed mean
Effective rank of the aggregate
effective rank of the aggregate (mean over layers)
36.3
37.9
37.0
34.5
27.1
20.8
49.6
35.6
36.2
37.9
38.0
38.0
40.8
41.0
37.9
33.0
27.7
26.8
0
10
20
30
40
50
mean
median
trimmed mean